

AI Design of Quantum Processors

Components 1–3 — from classical and quantum data to a trained classical→quantum model

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CONTEXT

We test whether classical data can predict quantum properties — first on the exactly-solvable oscillator, now on a real fluxonium qubit.

COMPONENT 1

Classical data

Phase-space orbits & energies — the model
INPUTS



COMPONENT 2

Quantum data

Spectra and Wigner functions; the $\langle \hat{\phi} \rangle, \langle \hat{n} \rangle$
trajectories are the model TARGETS



COMPONENT 3

Machine learning

Trained, benchmarked, and measured against
a physical axis

Central question: how much of a quantum system can be predicted from classical information alone — and where does that prediction break down?

Why this system?

The harmonic oscillator is exactly solvable in both worlds — every number has a built-in answer key.

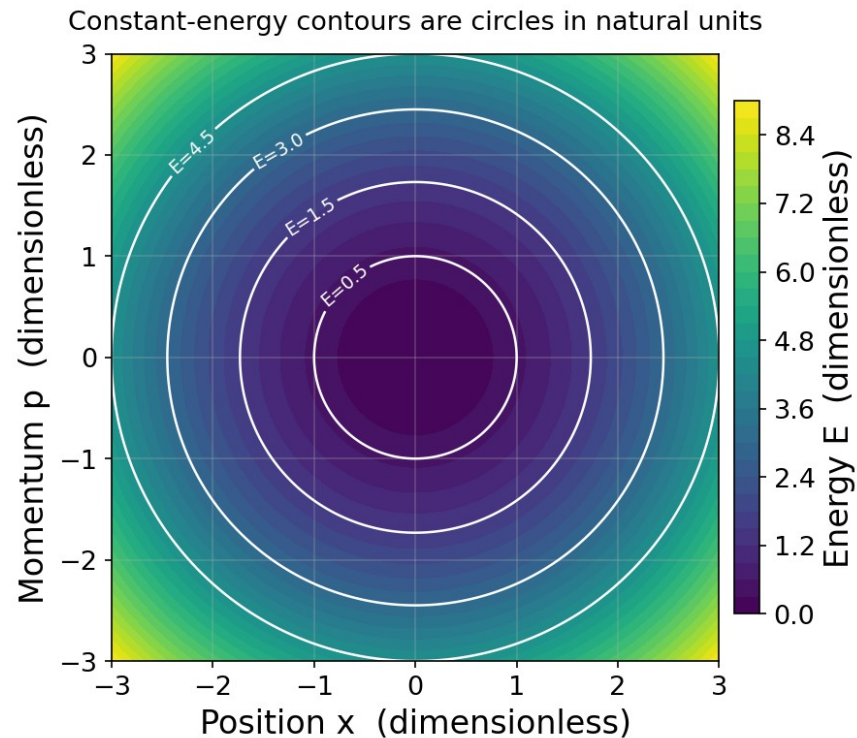
RESULTS · PART 1 OF 3

Component 1 — Classical data

The cheap side of the pipeline: phase-space orbits and energies for the harmonic, anharmonic and coupled oscillators — the model INPUTS. Every result is checked against an exact formula.

Constant-energy contours are closed elliptical orbits in phase space.

$$E(x, p) = p^2/2m + \frac{1}{2} m\omega^2 x^2 \quad \text{— the classical energy (Task 1a)}$$

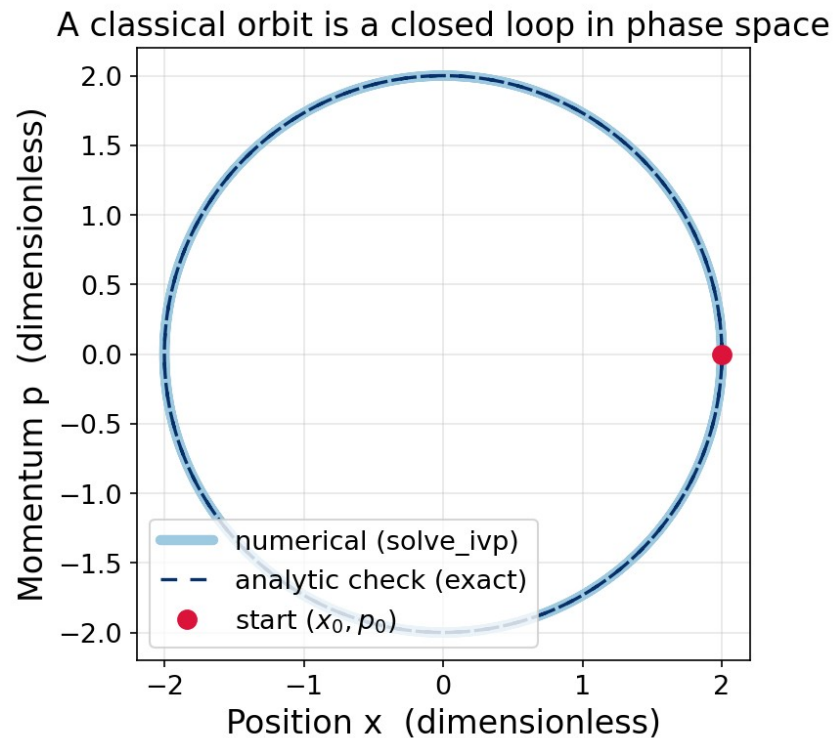


WHAT IT SHOWS

- Colour is total energy $E(x,p)$; white rings are constant-energy curves.
- Each ring is one allowed classical orbit — the system is locked onto it.
- In natural units the ellipses become circles; larger energy = larger ring.
- This map defines the classical states the ML model draws its inputs from.

Numerical trajectories match the exact solution and conserve energy.

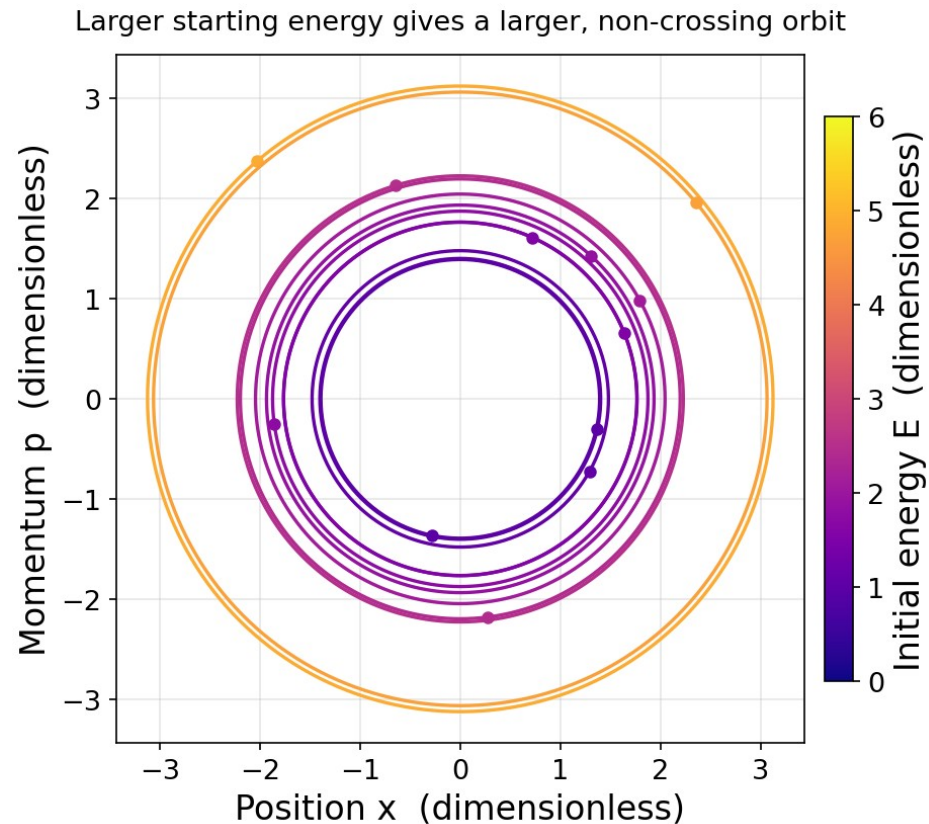
$$\dot{x} = \partial H / \partial p = p/m \quad \dot{p} = -\partial H / \partial x = -m\omega^2 x \quad \text{— Hamilton's equations (Task 1b)}$$



WHAT IT SHOWS

- solve_ivp steps Hamilton's equations forward in time to trace the orbit.
- The numerical path overlaps the analytic solution — the solver is verified.
- The orbit closes on itself and energy drift is $1.9\text{e-}9$ over four periods.
- Confidence here is what transfers to systems with no exact answer later.

Higher energy gives larger, non-crossing orbits — the classical feature set.

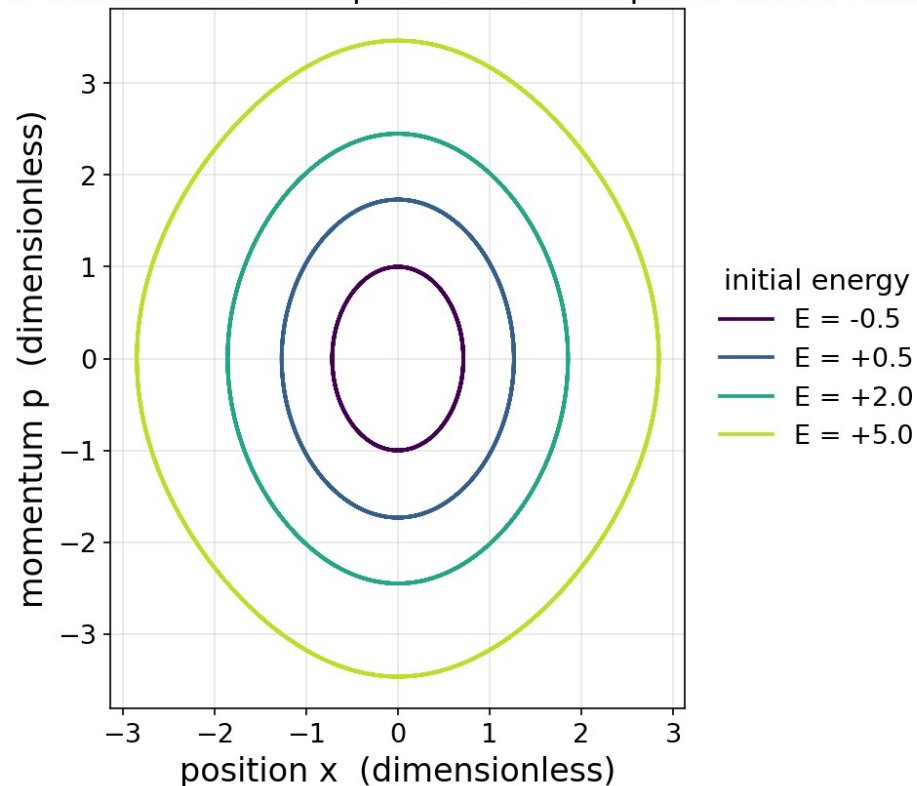


WHAT IT SHOWS

- Twelve random initial conditions, each coloured by its conserved energy.
- Orbits never intersect: a unique state has a unique future.
- Saved as `classical_trajectories.npy` — the inputs for Component 3.
- This nested structure is the 'feature space' the ML model learns from.

Adding a cosine well bends the orbits out of ellipses and ties the period to energy.

The cosine term bends the ellipses into non-elliptical closed orbits

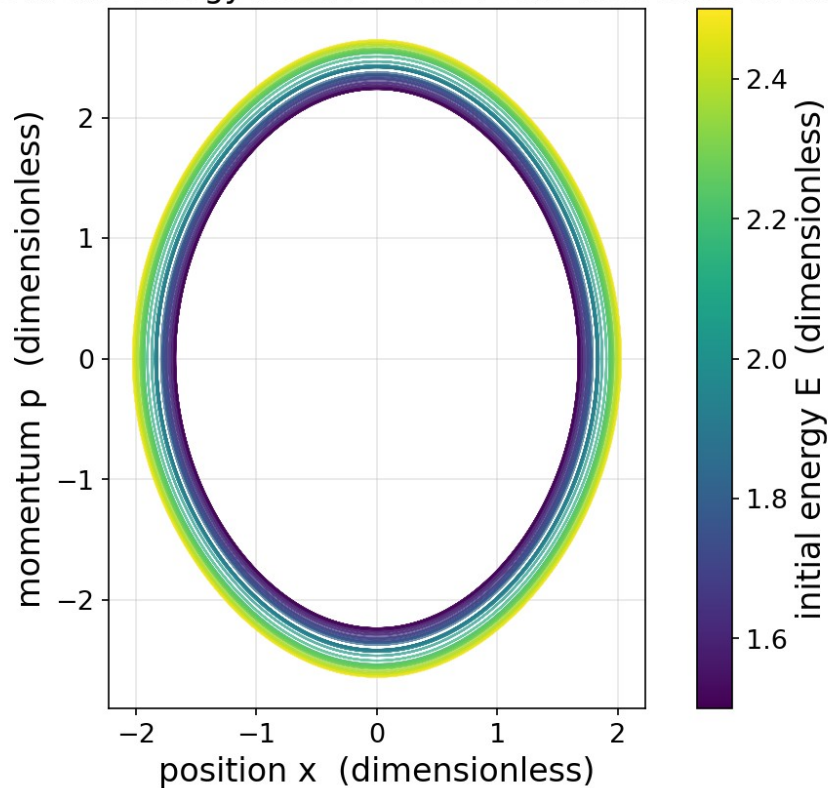


WHAT IT SHOWS

- $H = p^2/2m + \frac{1}{2}m\omega^2x^2 - V_0\cos(kx)$ — the classical shadow of a Josephson junction.
- Low-energy orbits still look elliptical; higher-energy orbits visibly deform.
- Anharmonicity is what makes a qubit addressable — evenly spaced levels cannot be driven two at a time.
- These are the classical inputs Component 3 pairs with fluxonium targets.

Nearby energies trace nearby orbits — the smooth structure the ML model later relies on.

Trajectories in the energy band $E = 2.0 \pm 0.5$ nest without crossing

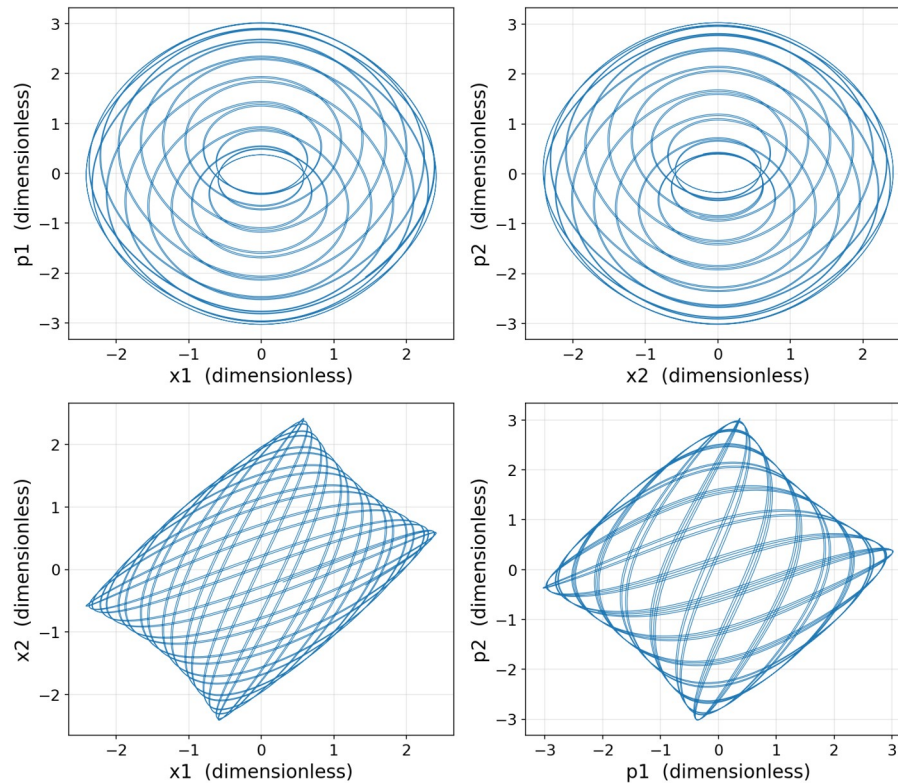


WHAT IT SHOWS

- Trajectories launched in a narrow energy band, $E = 2.0 \pm 0.5$, coloured by initial energy.
- The orbits nest without ever crossing: a small change in energy gives a small change in orbit.
- That is the property Component 3 depends on. If nearby classical inputs gave wildly different orbits, no smooth map from classical to quantum could exist.
- It also shows the motion is regular here — crossing orbits at the same energy would signal chaos.

Two coupled oscillators live in a 4-D phase space, so we read them through 2-D projections.

Four 2-D projections of the 4-D phase space ($E = 3.0$)

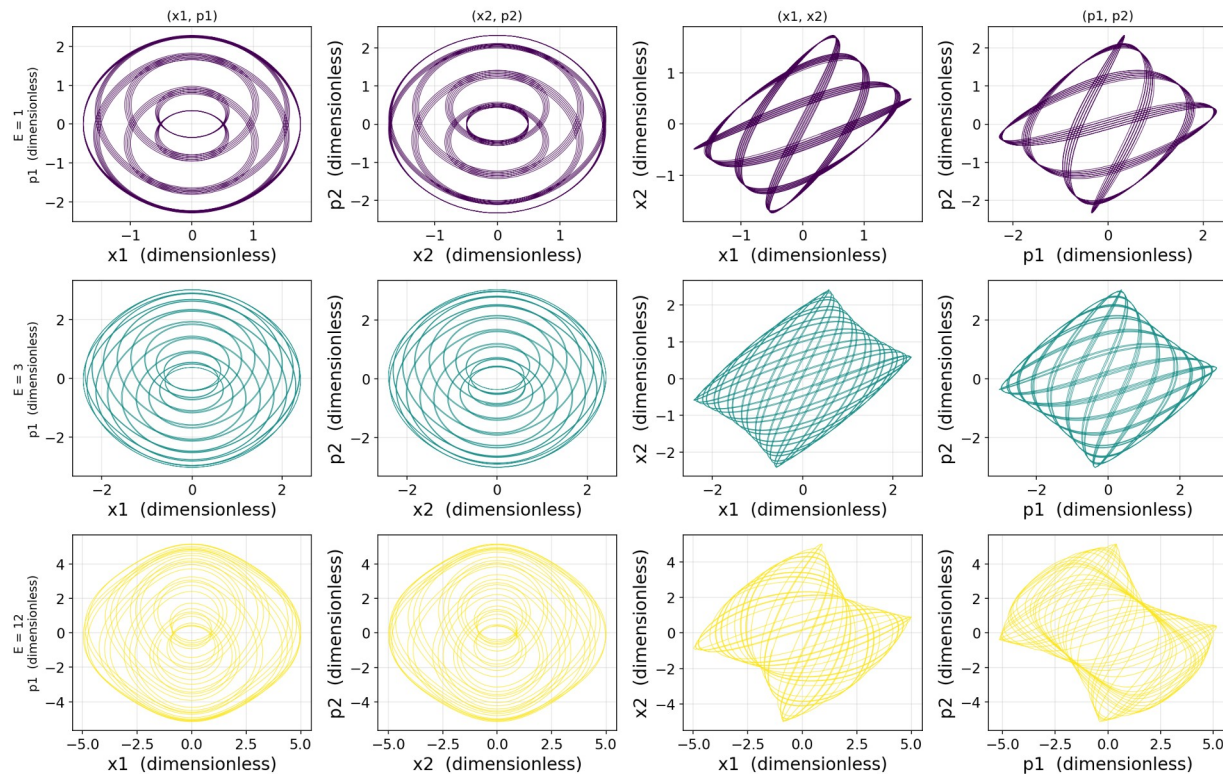


WHAT IT SHOWS

- Two cosine oscillators joined by momentum coupling $\lambda p_1 p_2$ with $\lambda = 0.3$.
- The state is (x_1, p_1, x_2, p_2) — four numbers, so the motion cannot be drawn directly.
- Each panel projects that 4-D motion onto one pair of coordinates.
- The next slide repeats all four panels at three energies.

Raising the energy grows the accessible region without ever disordering it.

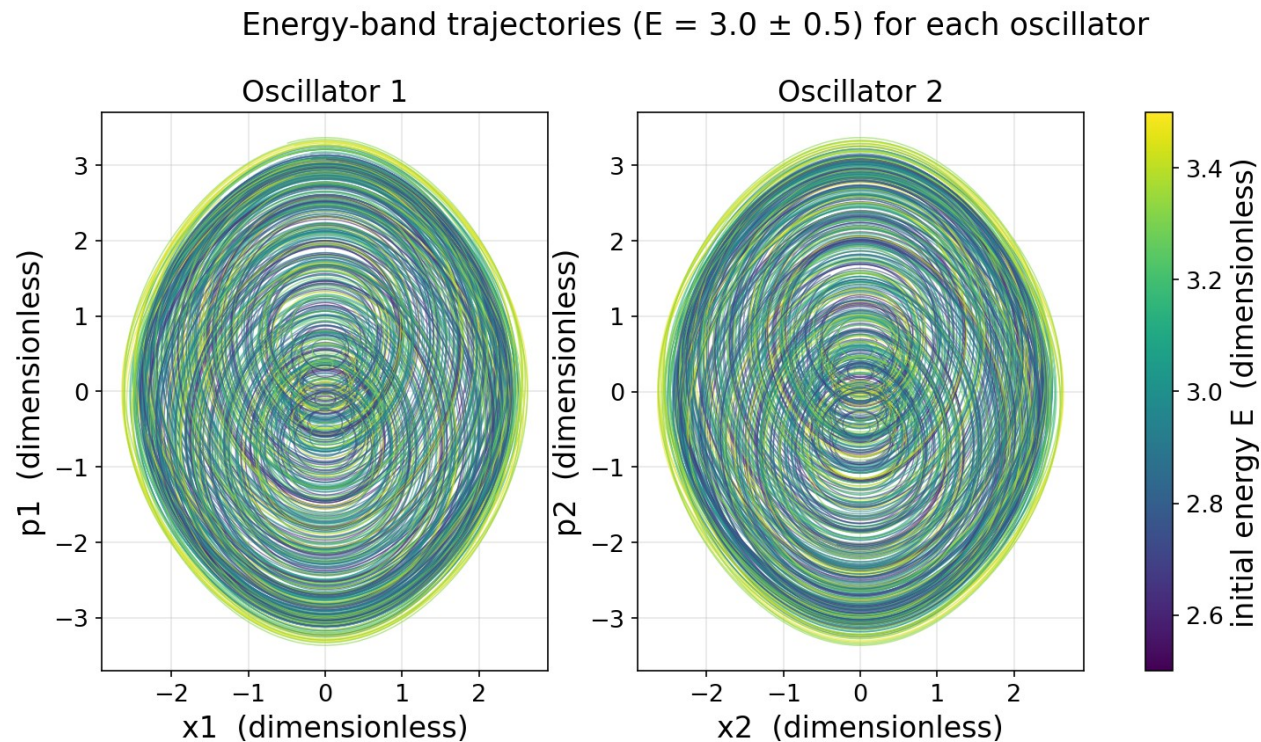
The four 2-D projections at three energies: the orbit fills a larger region as E grows, but stays on smooth nested curves at every energy



WHAT IT SHOWS

- The same four projections at $E_0 = 1, 3$ and 12 — twelve panels, one row per energy.
- The accessible region grows with energy, fastest in (p_1, p_2) , because the coupling acts through the momenta.
- The curves stay smooth and nested at every energy. Nothing scatters.
- This is the first evidence against my expectation of chaos at high energy — slide 12 measures it properly.

Each oscillator separately fills a band, and the coupling keeps the two correlated.

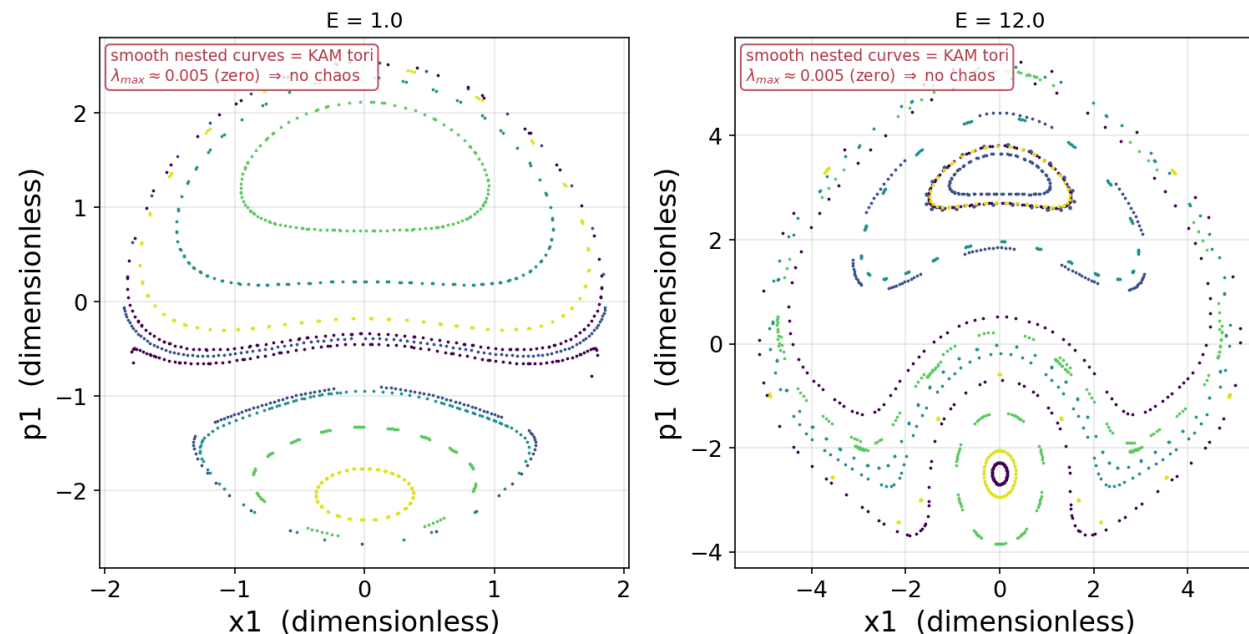


WHAT IT SHOWS

- The (x_1, p_1) and (x_2, p_2) projections for a band of initial energies, $E = 3.0 \pm 0.5$.
- Each oscillator explores a filled region whose size grows with its share of the energy.
- The two are not independent: the momentum coupling exchanges energy between them, so their regions stay correlated.
- Same message as Task 3(c), now with two degrees of freedom — the structure survives coupling.

I expected chaos at high energy. Measuring it properly says the motion stays regular.

Poincaré sections ($x_2 = 0, dx_2/dt > 0$): regular motion at both energies



WHAT IT SHOWS

- A Poincaré section records the state only as it crosses $x_2 = 0$ in one direction.
- Smooth nested curves mean regular, quasiperiodic motion (KAM tori); a scattered cloud would mean chaos.
- Both $E = 1$ and $E = 12$ give nested curves. The Lyapunov exponent confirms it: $\lambda \approx 0.004\text{--}0.007$, zero to within the log t/t floor.
- The nonlinearity is a bounded cosine while the harmonic term grows without limit — so higher energy makes this system MORE nearly integrable.

RESULTS · PART 2 OF 3

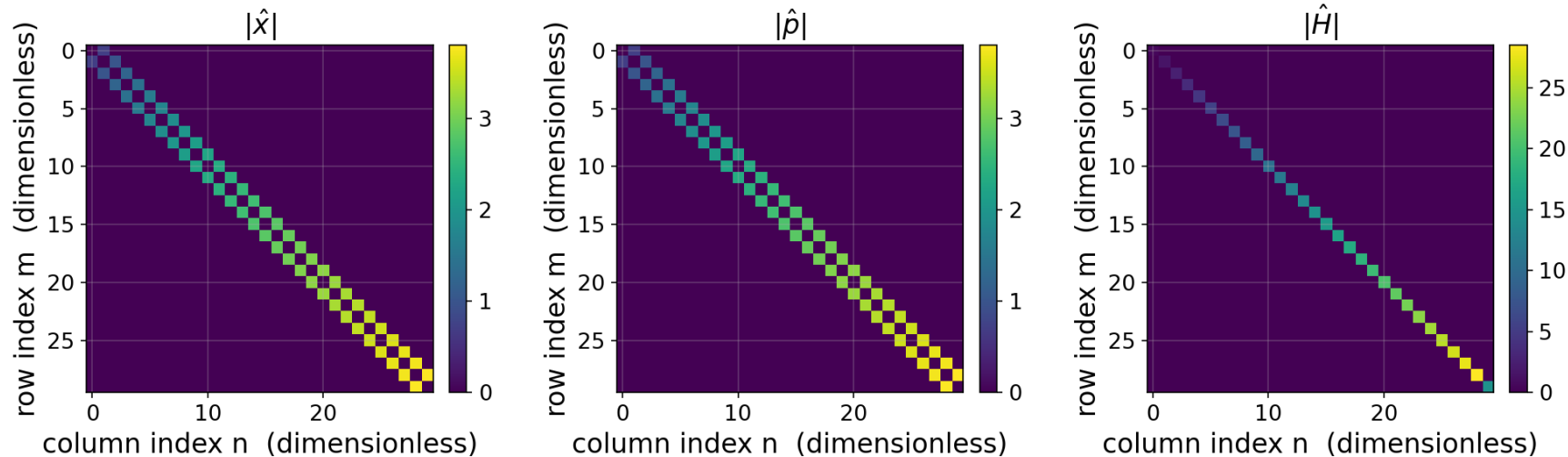
Component 2 — Quantum data

The expensive side: operators, spectra, Wigner functions and wave-packet dynamics — from the textbook oscillator to a real fluxonium qubit. These are the model TARGETS.

The quantum operators reveal the ladder structure of the oscillator.

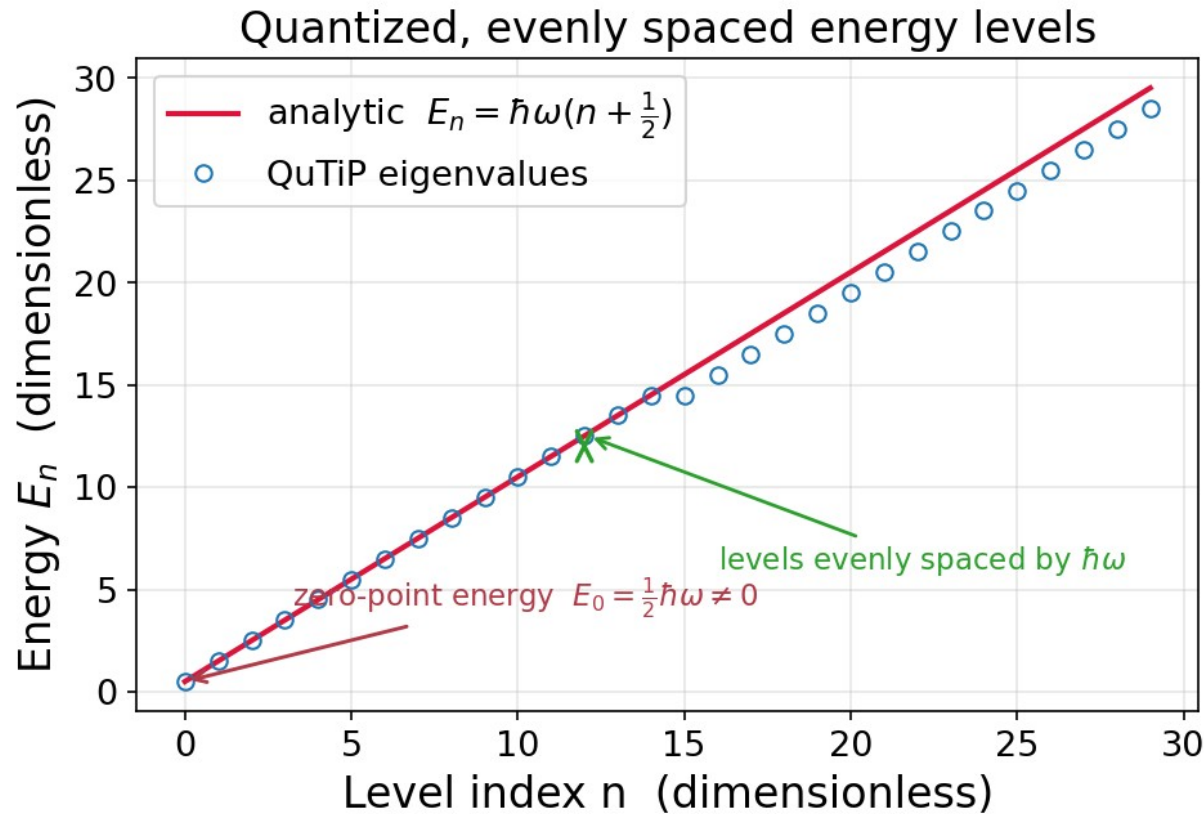
$$\hat{x} = \sqrt{(\hbar/2m\omega)}(\hat{a} + \hat{a}^\dagger) \quad \hat{p} = -i\sqrt{(\hbar m\omega/2)}(\hat{a} - \hat{a}^\dagger) \quad \hat{H} = \hat{p}^2/2m + \frac{1}{2} m\omega^2 \hat{x}^2$$

Operator matrices in the energy (Fock) basis



Takeaway: $\hat{H}$ is diagonal (each level has a definite energy), while $\hat{x}$ and $\hat{p}$ have only off-diagonal entries — they connect a level to its immediate neighbours. That off-diagonal stripe is the ladder, drawn.

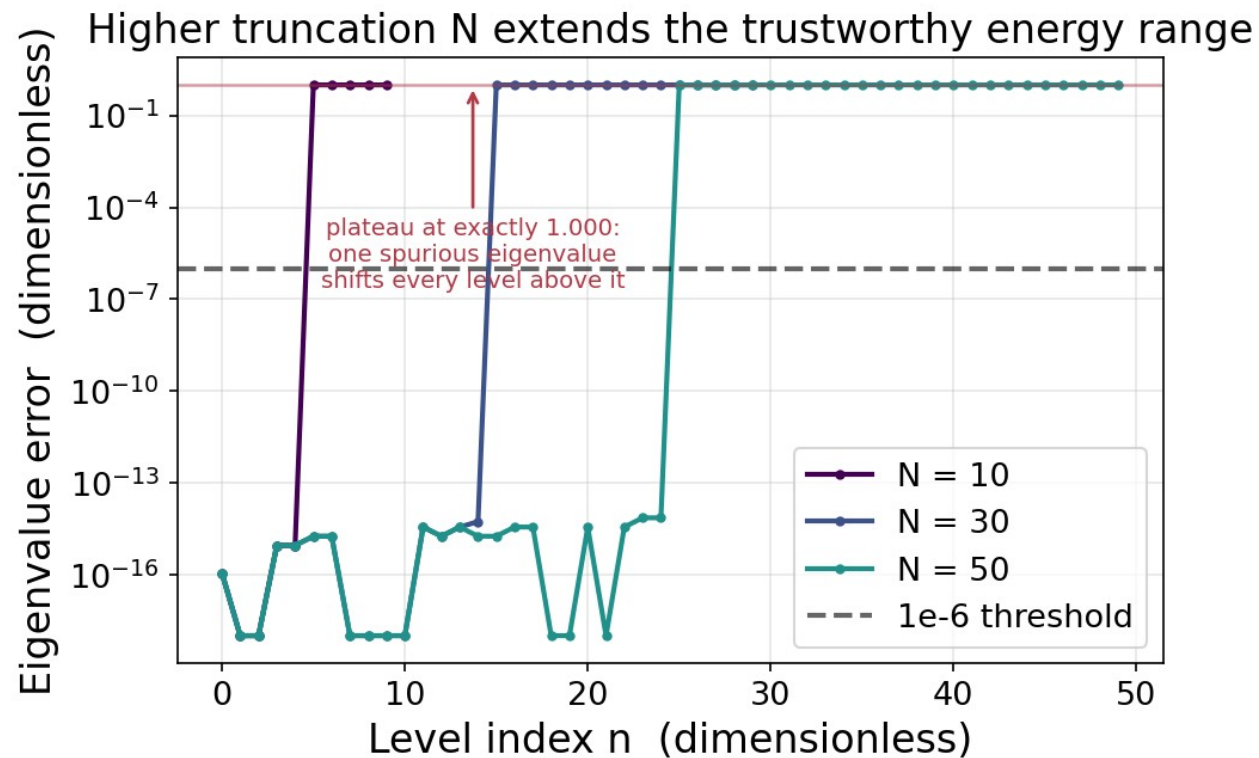
Quantum energy is discrete, evenly spaced, with a nonzero zero-point energy.



WHAT IT SHOWS

- QuTiP eigenvalues land exactly on the analytic line $E_n = \hbar\omega(n + \frac{1}{2})$.
- Ground state $E_0 = \frac{1}{2}\hbar\omega \neq 0$ — a purely quantum effect.
- Levels are equally spaced by $\hbar\omega$ (the ladder).
- High- n points drift: the expected signature of finite truncation.

The truncation level N sets how many energy levels are trustworthy.

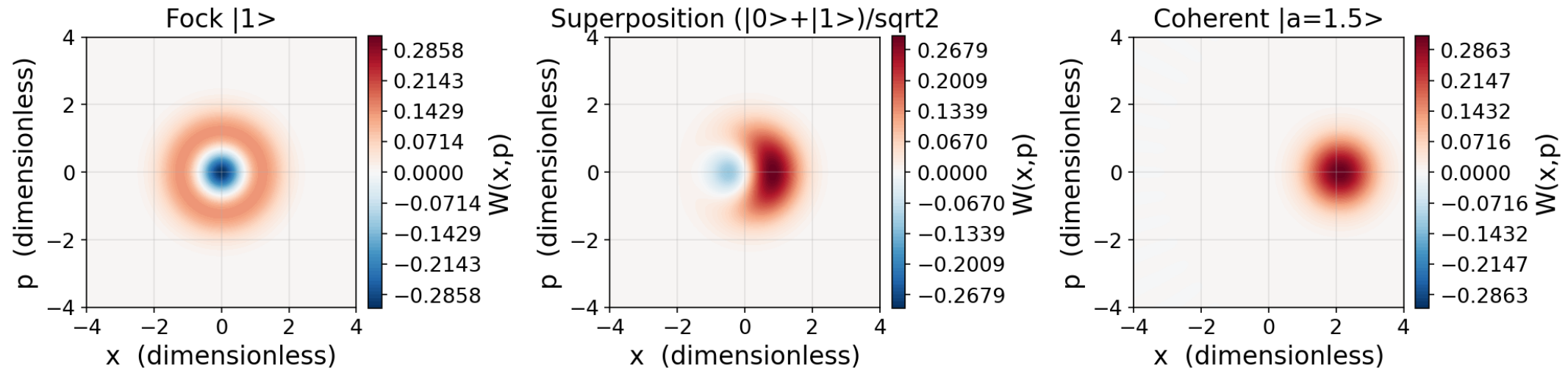


WHAT IT SHOWS

- Error sits at machine-zero, then jumps to a flat plateau at exactly 1.000 — not a growing drift.
- Cause: truncation makes one spurious eigenvalue at $(N-1)/2$; sorting then shifts every level above it by one rung.
- So the lowest $\lceil (N-1)/2 \rceil$ levels are exact: 5 of 10, 15 of 30, 25 of 50.
- Rule: raise N until the quantity of interest stops changing.

Wigner functions expose non-classicality through negativity and interference.

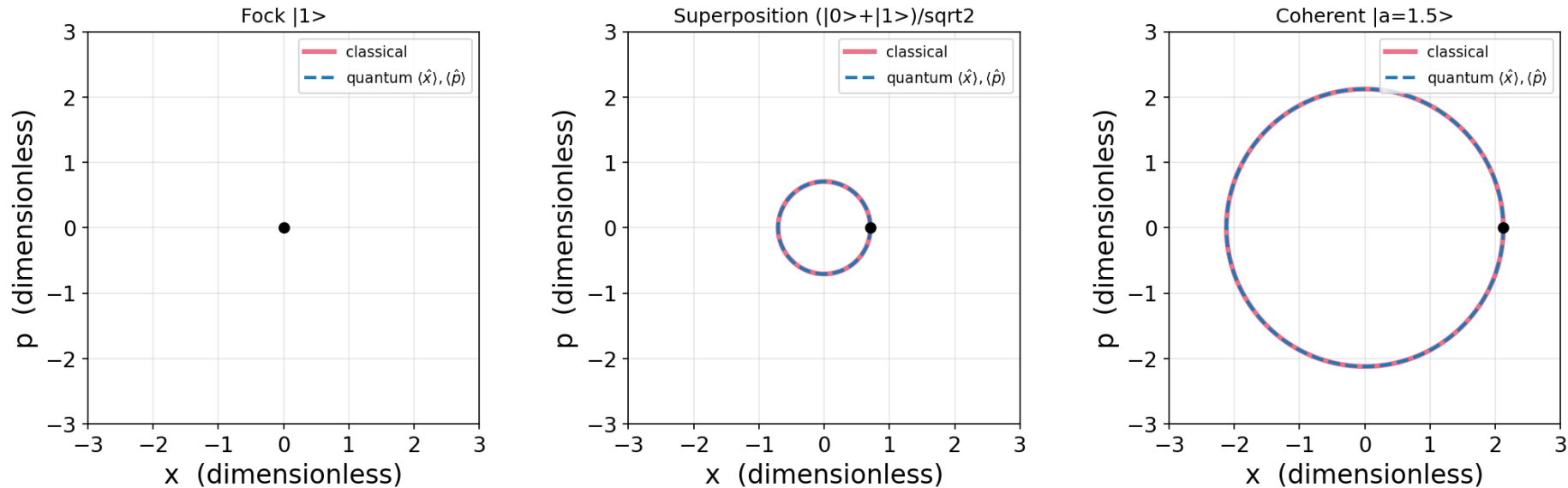
Wigner functions at $t = 0$ (blue = negative = non-classical)



Takeaway: The Fock state has a negative core and the superposition shows interference fringes (both impossible classically); the coherent state stays positive — the most classical state. Task 2(c) animates all three at a fixed colour scale — [movies/wigner_{coherent,fock,superposition}.gif](#), playable from the repo.

Quantum averages trace the classical orbit, but averaging hides the spread.

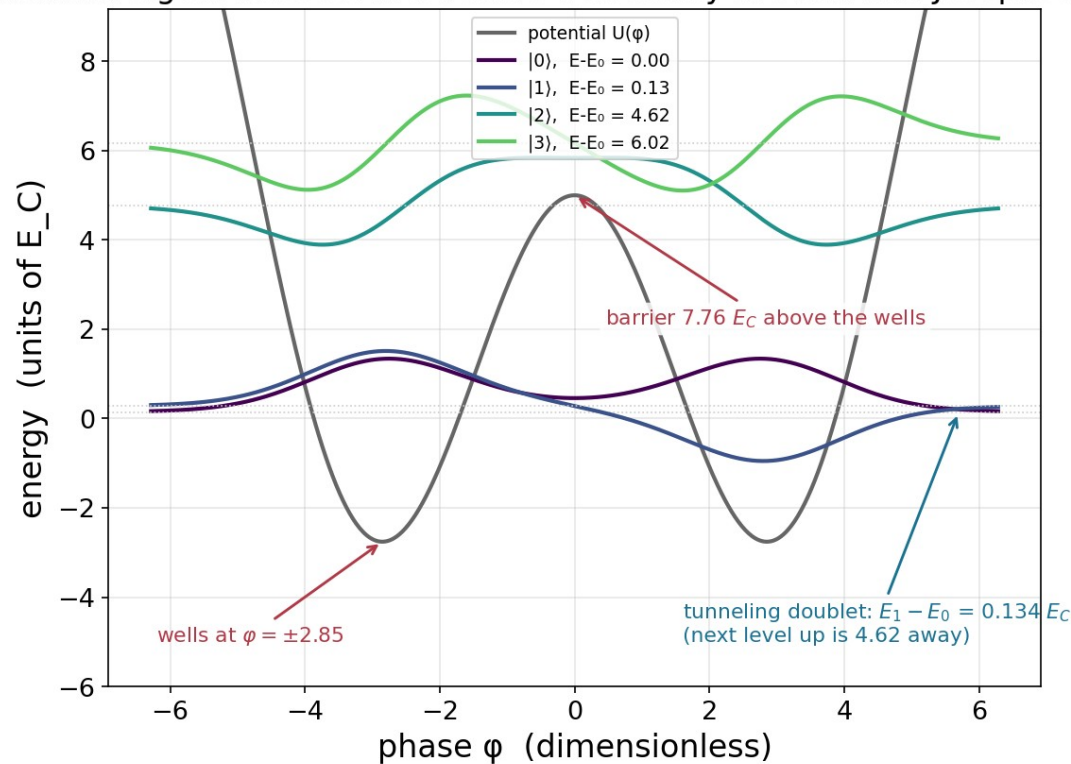
Quantum averages vs. classical orbits



Takeaway: For the coherent state $\langle \hat{x} \rangle, \langle \hat{p} \rangle$ follow the classical circle exactly (Ehrenfest's theorem); the Fock state sits at the origin. The averages discard the variance and interference the Wigner picture keeps.

At half flux the fluxonium is a double well whose two lowest states form a tunneling doublet.

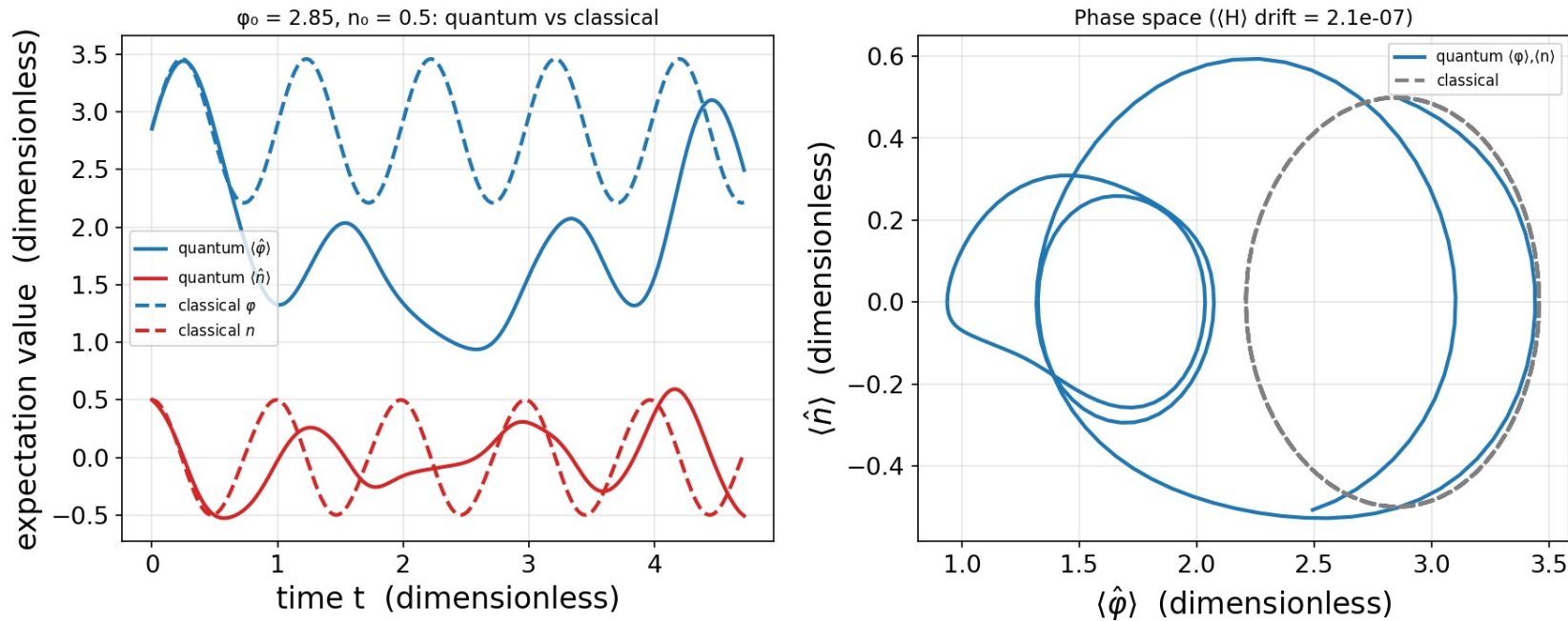
Fluxonium eigenstates sit in a double well set by the flux and Josephson terms



WHAT IT SHOWS

- $\hat{H} = 4E_C \hat{n}^2 + \frac{1}{2}E_L \hat{\phi}^2 - E_J \cos(\hat{\phi} + \phi_{\text{ext}})$ — scqubits' convention, at $E_J/E_C = 5$, $E_L/E_C = 0.5$.
- $\phi_{\text{ext}} = \pi$ is the symmetric sweet spot: wells at $\phi = \pm 2.85$ with a barrier at $\phi = 0$, $7.76 E_C$ above them.
- $E_1 - E_0 = 0.134 E_C$ while $E_2 - E_0 = 4.62 E_C$ — a 34× gap that isolates a clean two-level system.
- That tiny splitting is tunneling through the barrier, and it has no classical counterpart.

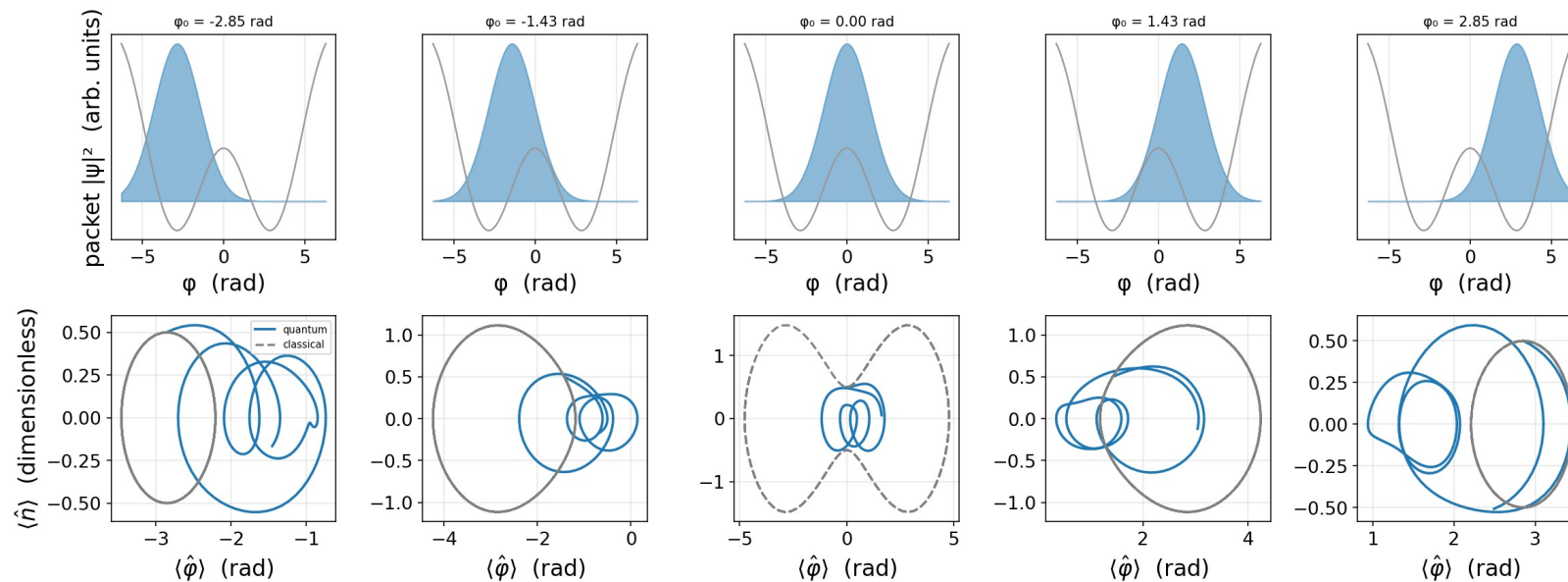
A wave packet tracks the classical orbit for half a period, then falls behind it.



Takeaway: Both start at the well minimum $\phi_0 = 2.85$ with the same charge kick $n_0 = 0.5$. They agree until $t \approx 0.7$, then the quantum average loses amplitude. The reason is precise: Ehrenfest gives $d\langle \hat{n} \rangle / dt = \langle -\partial U / \partial \phi \rangle$, the average of the force — which equals the classical force at the average position only for a harmonic potential. $\langle \hat{H} \rangle$ is conserved to $2.1e-7$, so this is physics, not solver error.

Launched on the barrier, the classical and quantum pictures stop describing the same system.

Fluxonium wave packets across the wells: quantum averages vs classical trajectories



Takeaway: Five starting phases from one well, across the barrier, to the other. In the wells ($\phi_0 = \pm 2.85$) the two come closest. On the barrier top ($\phi_0 = 0$) the classical point sweeps a wide figure-eight through both wells while the packet splits and stays put — the same initial condition, two completely different answers. That gap is what Component 3 has to learn.

RESULTS · PART 3 OF 3 — NEW THIS UPDATE

Component 3 — Machine learning

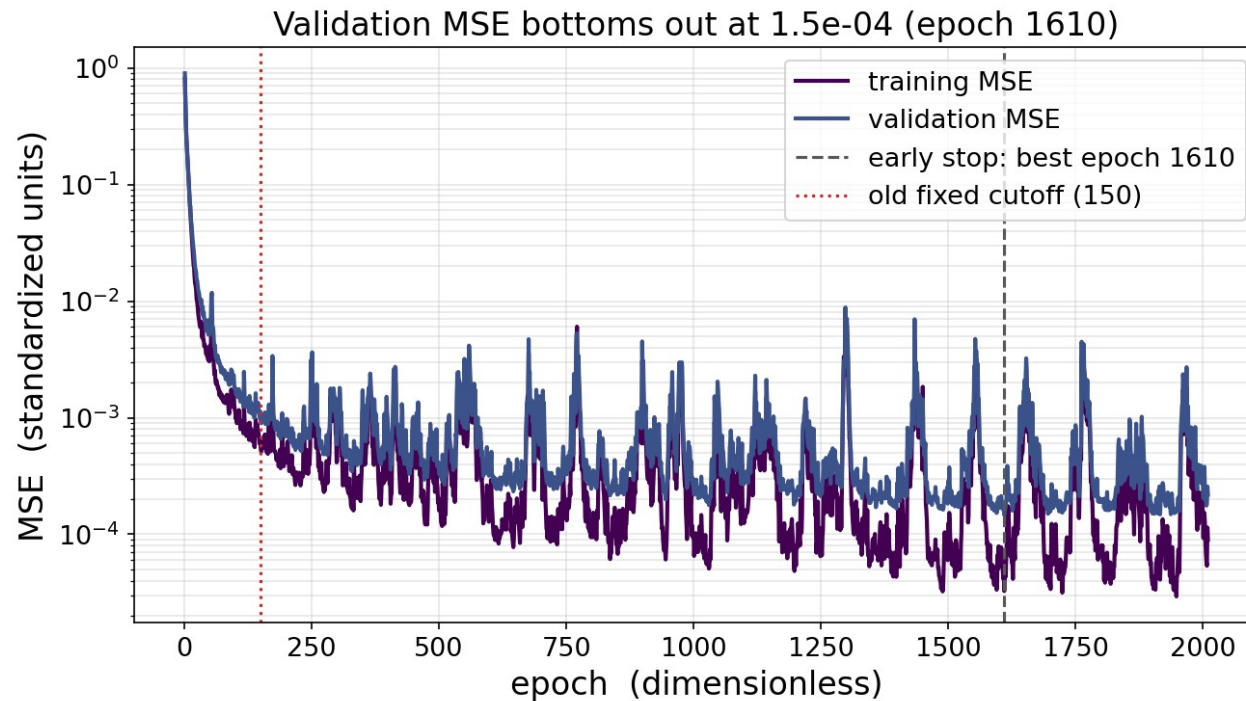
Trained to early stopping, scored against three honest baselines, and measured against a physical axis — the classical→quantum breakdown is now located.

Training to the validation minimum instead of a fixed 150 epochs improved the model 6.4-fold.

$$h_1 = \text{ReLU}(W_1 A + b_1) \quad h_2 = \text{ReLU}(W_2 h_1 + b_2) \quad \hat{B} = W_3 h_2 + b_3$$

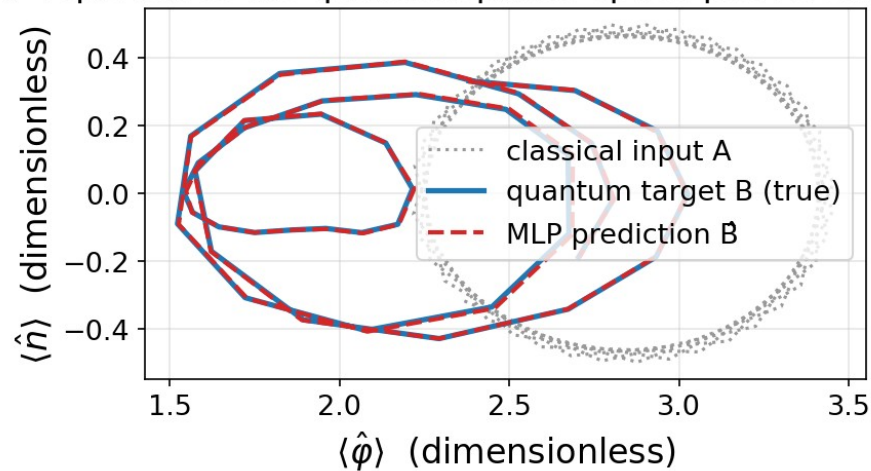
WHAT IT SHOWS

- 300 paired trajectories: classical solve_ivp input A and fluxonium sesolve target B from the same packet.
- Two hidden layers of 128 with ReLU, MSE loss, Adam at lr = 1e-3, an 80/20 split.
- Early stopping keeps the weights at the validation minimum — epoch 1610, MSE 1.47e-4.
- The old fixed cutoff (red dotted) stopped while validation was still falling: 9.3e-4, 6.4× worse.



On held-out data the predicted quantum path follows the true one, not the classical input.

The MLP reproduces the quantum phase-space path from classical input



WHAT IT SHOWS

- One validation trajectory, un-standardized back into physical units.
- Grey dotted is the classical input A; blue is the true quantum target B; red dashed is the prediction.
- The prediction sits on the quantum curve, so the network learned the correction — not a copy of its input.
- Physically consistent pairs are what make this learnable — mismatched pairs did 9× worse.

The MLP beats all three honest baselines — but linear regression is the one that matters.

1.067

COPY-CLASSICAL

hand back the classical input
unchanged

0.077

K-NN (K = 1)

look up the closest training trajectory

0.026

LINEAR REGRESSION

best straight-line map, solved exactly

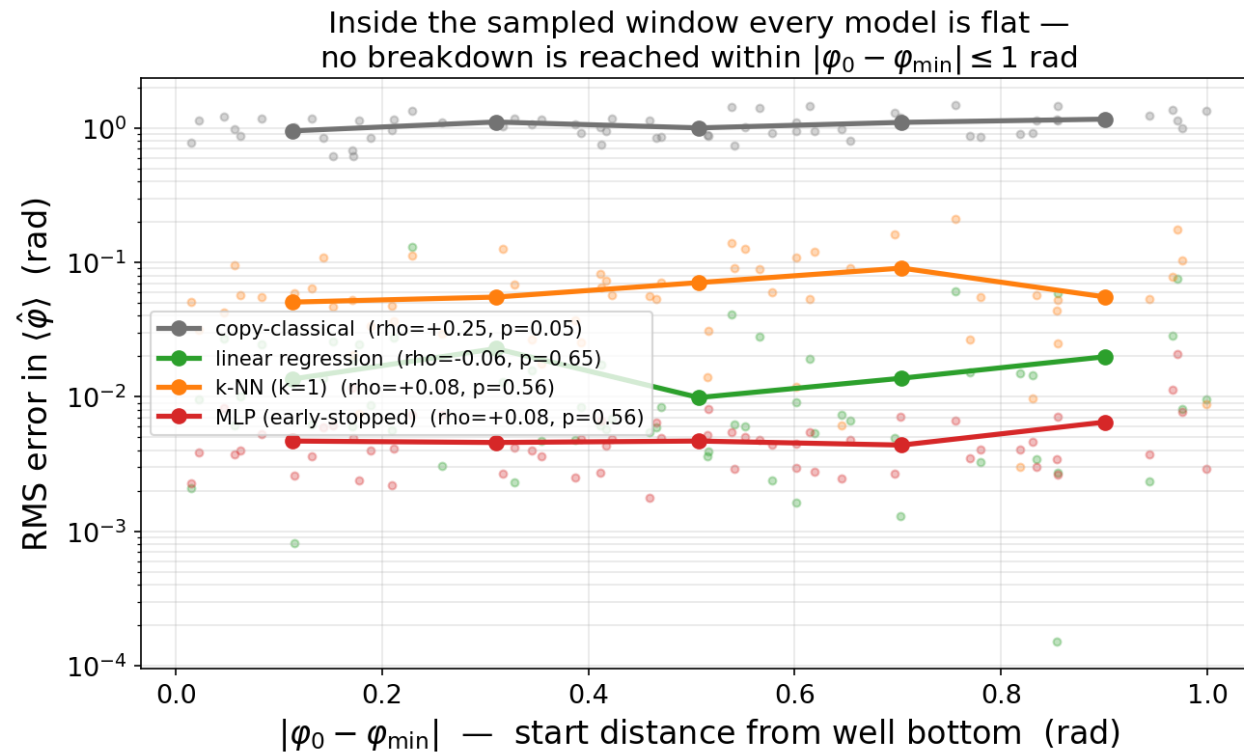
0.0057

MLP

the trained network

Takeaway: RMS error in $\langle \hat{\phi} \rangle$ (radians) on the same 60 held-out trajectories. Beating copy-classical 186× shows the network reconstructs the quantum correction rather than passing its input through. But inside one well the motion is near-harmonic, and the harmonic classical→quantum map genuinely is linear — so a least-squares fit with no training loop already reaches 0.026. The honest headline is 4.5× over linear regression, not 186× over copying.

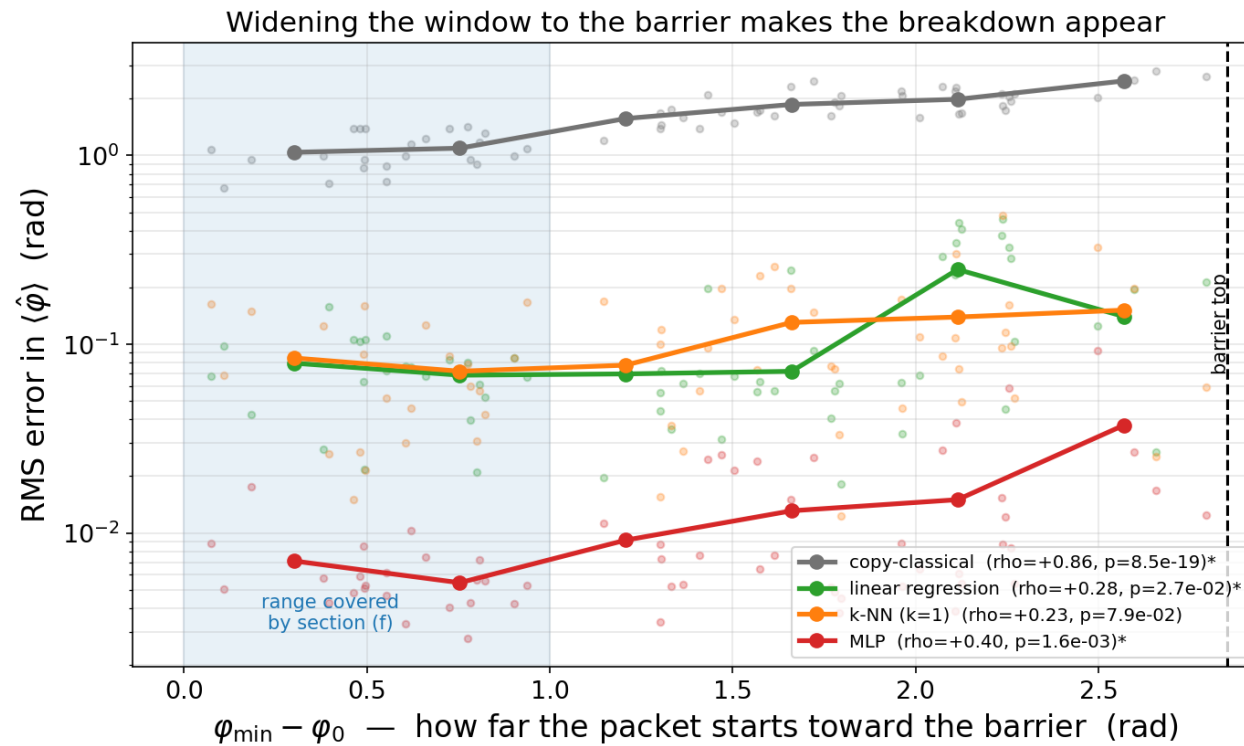
My first attempt to find the breakdown found nothing — because it could not have.



WHAT IT SHOWS

- Error against distance from the well bottom is flat for every model: Spearman $\rho = +0.08$, $p = 0.56$ for the MLP.
- The first draft of this figure was titled 'error grows with distance'. Measuring the trend said otherwise, so it was retitled.
- But the barrier sits 2.85 rad from the minimum and this dataset sampled only to 1.00 rad — 35% of the way.
- A null from a measurement that could not have detected the effect is not evidence of absence.

The breakdown sits exactly where Ehrenfest's condition fails — at the barrier.



WHAT IT SHOWS

- A second dataset, from the well bottom to the barrier top, on a monotone one-sided axis.
- MLP error climbs $5.2\times$ ($\rho = +0.40$, $p = 0.0016$); copy-classical climbs $2.4\times$ ($\rho = +0.86$, $p = 8.5e-19$).
- Copy-classical's rise IS the quantum correction growing — a direct measurement of where the two pictures part company.
- Truncation re-checked for the wider window: cutoff 80 vs 110 agree to $8.2e-7$ rad.

VERIFICATION

Checking every number against an exact formula is what caught all four corrections.

$$5 \times 10^{-15}$$

SPECTRUM

max error vs $E_n = \hbar\omega(n+\frac{1}{2})$, lowest 15 levels

$$1.9 \times 10^{-9}$$

ENERGY DRIFT

over four periods, at $\text{rtol} = \text{atol} = 1\text{e-}10$

$$8.2 \times 10^{-7}$$

TRUNCATION

cutoff 80 vs 110 across the full sampling window

Takeaway: Four corrections came from this discipline, none of which raised an exception. (1) The fluxonium mixed two coordinate conventions — fixing it improved validation MSE nine-fold. (2) A claimed order-to-chaos transition the Lyapunov exponent did not support. (3) A null result that was a limit of the sampling window, not a property of the map. (4) Two numbers in my own write-up that a re-run did not reproduce. A figure is a picture, not a measurement — and a number in prose is a claim until it is traced back to output.

What I don't yet understand, and what I'd do next.

WHAT I DON'T YET UNDERSTAND

- Is the model learning physics, or interpolating between 240 stored examples? Testing outside the training window would separate the two.
- Linear regression already reaches 0.026 rad, so most of the in-well map is trivially linear. How much genuine nonlinear structure is the network actually capturing?
- The MLP degrades fastest in relative terms toward the barrier (5.2×). Is that the map getting harder, or the training data thinning out there?
- Does a deeper well (larger E_J/E_C) bring classical and quantum back together, as the correspondence principle says it should?

WHAT I'D LIKE TO DO NEXT

- Train on the residual $B - A$ rather than B — the part Ehrenfest discards, and the only part that is not near-linear.
- Sweep across parameters (E_J/E_C , E_L/E_C , flux) rather than one point, which is where the Science 2022 generalization results apply.
- Predict a quantity with no classical analogue — the tunneling splitting — instead of a trajectory.
- Re-run Task 4 in the genuinely chaotic regime — $\lambda = 0.8$ with $V_0 = 8$ gives a Lyapunov exponent of 0.11 — if that is worth the time.

Which fluxonium property is actually worth predicting for real device design? — that one is genuinely your call.